

PolyGate

An Explainable, Policy-Driven AI Gateway

One governed request. Two user experiences. One observable cloud platform.

Cloud Computing

SWS3004_03

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BACKGROUND

Multi-provider applications must trade quality and privacy against cost, latency, and reliability. unified endpoint alone cannot govern those choices.



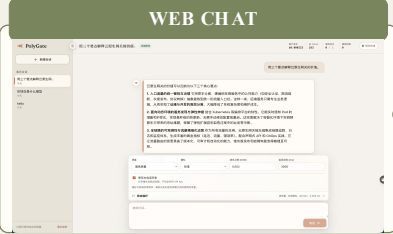
SCIENTIFIC QUESTION

How can a cloud-native AI gateway select, govern, execute, and explain workloads under simultaneous constraints?



USER LAYER

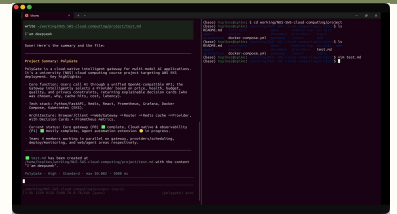
Web Chat turns preferences into routing constraints. Pi Agent sends model calls through PolyGate while tools stay local.



WEB CHAT

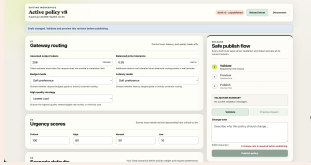


PI AGENT



ADMINISTRATOR CONTROL PLANE

POLICY MANAGEMENT



Versioned policies: Validate → Preview → Publish → Compare → Roll Back. Privacy and queue rules govern routing.



Validate



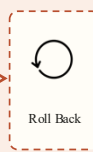
Preview



Publish



Compare



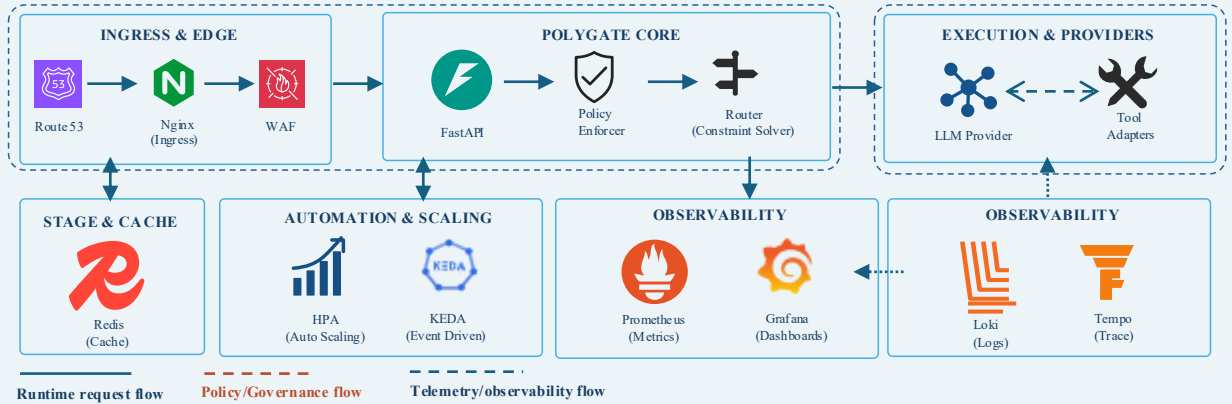
Roll Back

CLOUD-NATIVE SYSTEM

AWS EKS runs the Gateway, automation, cache, scaling, providers, and observability path.



aws | EKS | K8s



RESEARCH DESIGN

Separate interaction, governance, and execution into independently deployable layers.



METHOD & MECHANISM



Intent → Enforce → Route → Execute → Explain → Observe

KEY RESULTS

Explainable provider choice
Version-aware operations
Observable cost and health



INNOVATIONS

Constraint-first routing
Auditable policy lifecycle
Cloud-native automation



DISCUSSION

Exact caching and per-Pod circuit state define current reliability boundaries.



CONCLUSION & OUTLOOK

PolyGate makes multi-model access governable and observable. Next: shared state, durable stores, and stronger tenant control.

